

EDUVISION ANALYTICS

Higher Education Performance Dashboard

A Data Analytics & Visualization Project on Global University Rankings using QS and Times Higher Education (THE) Datasets

PROJECT DOCUMENTATION

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ABOUT THIS DOCUMENTATION

This document presents the complete project documentation for EduVision Analytics, a Higher Education Performance Dashboard built by integrating and analyzing QS and Times Higher Education (THE) university ranking datasets. It covers the project background, objectives, dataset description, the Python-based data cleaning workflow, the KPI and analytics methodology, and a detailed explanation of the five Tableau dashboards — Home, University, Research, Student, and Country Comparison — along with the project's conclusion, future scope, and GitHub repository.

Project Repository (GitHub):
github.com/springboardmentorpl/Higher-Education-Performance-Dashboard-Group-1/tree/Unni_Krishnan_R

1. Project Overview

EduVision Analytics is a data analytics and visualization project developed to study and compare the performance of universities across the world using globally recognized higher-education ranking data. The project brings together data engineering and interactive dashboard design to convert raw ranking data into insights that are easy to interpret and explore.

1.1 Background

Higher-education ranking agencies such as QS (Quacquarelli Symonds) and Times Higher Education (THE) publish yearly university performance data based on indicators including academic reputation, employer reputation, citations, teaching quality, research strength, and international outlook. These datasets are large, differ in structure between sources, and are difficult to interpret directly in raw spreadsheet form. EduVision Analytics addresses this by cleaning, standardizing, and combining the QS and THE datasets into a single analytical structure and presenting the findings through interactive Tableau dashboards.

1.2 Purpose of EduVision Analytics

The purpose of the project is to give students, researchers, and academic administrators a single, interactive platform to explore university performance from multiple perspectives — overall standing, research strength, student experience, and country-level comparison — instead of relying on static ranking tables published separately by each agency.

1.3 Why Higher-Education Performance Analysis is Useful

- Helps prospective students compare universities on academic reputation, research output and student experience.
- Allows institutions to benchmark themselves against peer universities and identify improvement areas.
- Supports policy makers and education bodies in understanding country-wise higher-education performance.
- Converts multi-source ranking data (QS and THE) into a consistent, analysis-ready format.
- Provides an interactive alternative to static PDF/HTML ranking tables published by ranking agencies.

1.4 Project Summary

EduVision Analytics follows a complete analytics pipeline — starting with QS and THE ranking data, moving through Python-based cleaning and preprocessing in Jupyter Notebook, and ending with a five-dashboard Tableau solution (Home, University, Research, Student, and Country Comparison). Each dashboard focuses on a specific dimension of higher-education performance, supported by KPI cards, comparative charts, and geographic visualizations, enabling users to explore the data interactively rather than reading static reports.

1.5 Scope and Intended Users

- Covers the end-to-end analytics lifecycle: data collection, integration, cleaning, KPI design and dashboard development.
- Focused on descriptive and comparative analytics of current rankings rather than predictive modelling.
- Delivered as a five-dashboard Tableau suite supported by this written project documentation.
- Intended for students and researchers comparing universities, and institutions benchmarking against global peers.

2. Problem Statement & Objectives

2.1 Problem Statement

University ranking information published by QS and Times Higher Education is spread across separate datasets that use different field names, scoring scales, and formats. Comparing universities across both sources, or analyzing performance by country, research strength, or student experience, requires manual cross-referencing that is time-consuming and error-prone. There is a need for a unified, cleaned dataset and an interactive visualization system that allows performance to be explored dashboard-wise rather than through raw spreadsheets.

2.2 Key Challenges

- Combining two ranking datasets (QS and THE) that differ in structure, field naming, and score scales.
- Handling missing values present across ranking and score fields without distorting the analysis.
- Standardizing rank and score fields into consistent numeric formats (e.g. Rank_Numeric, Rank_Normalized, Overall_Score_Numeric).
- Standardizing free-text university and country name variations for accurate grouping and comparison.
- Designing multiple Tableau dashboards that stay readable while presenting many KPIs and chart types together.
- Ensuring dashboard navigation and filters behave consistently across the Home, University, Research, Student and Country Comparison views.

2.3 Project Objectives

- To collect and integrate QS and THE higher-education ranking datasets into a single analytical dataset.
- To clean and preprocess the combined data using Python in a Jupyter Notebook environment.
- To standardize ranking and score fields for accurate cross-source comparison.
- To design and build five interactive Tableau dashboards covering overall performance, university-level, research, student, and country-level analysis.
- To present key performance indicators (KPIs) that summarize academic reputation, citations, research and student-related measures.
- To enable comparative analysis of universities and countries through charts, maps and tables.
- To document the complete workflow from data source to final dashboard for academic submission.

2.4 Expected Outcome

A cleaned, standardized higher-education dataset and a five-dashboard Tableau analytics suite (EduVision Analytics) that allows interactive exploration of global university performance by overall score, research strength, student experience, and country, supported by KPI cards, comparative charts, and a global outlook map.

3. Data Sources & Dataset Description

3.1 Data Sources

The project uses two globally recognized higher-education ranking datasets as its primary data sources:

- **QS World University Rankings** — published by Quacquarelli Symonds; includes indicators such as academic reputation, employer reputation, faculty/student ratio, citations, and international outlook.
- **Times Higher Education (THE) World University Rankings** — includes indicators such as teaching, research environment, citations, industry income, and international outlook.

3.2 Important Fields Used in the Project

Field	Description
University	Name of the university (standardized across QS and THE).
Country	Country/territory of the university (standardized text values).
Rank_Numeric	Cleaned numeric form of the published rank.
Rank_Normalized	Rank value normalized for cross-source comparison.
Overall_Score_Numeric	Standardized numeric overall performance score.
Source	Indicates whether the record originated from QS or THE.
Research / Citation Fields	Indicators covering research output, citations, and research environment.
Student / Teaching Fields	Indicators covering faculty support, teaching strength, and student experience.

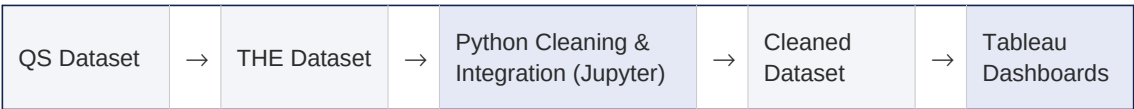
3.3 Data Integration

Since QS and THE publish their rankings independently, the project integrates both datasets by aligning common fields such as university name, country, rank, and overall score. Source-specific fields are retained alongside a common **Source** field so that records can still be compared or filtered by the originating ranking agency inside the Tableau dashboards (for example, in the Research dashboard's Source Insights and Research vs Citations views).

3.4 Dataset Characteristics

- Covers universities across a wide range of countries and regions.
- Combines ranking, score, research, and student-related indicators from two independent sources.
- Contains both numeric fields (ranks, scores) and categorical/text fields (university, country).

3.5 Data Flow Overview



4. Data Collection & Cleaning

All data preparation for EduVision Analytics was carried out in Python using a Jupyter Notebook, prior to importing the cleaned dataset into Tableau. The preprocessing workflow followed the stages below.

- 1. Data Collection** — The QS and Times Higher Education (THE) ranking datasets were collected as the raw source files for the project.
- 2. Loading Datasets** — Both datasets were loaded into a Jupyter Notebook using Python for inspection and preprocessing.
- 3. Source Identification** — A Source field was introduced/retained to identify whether each record originated from QS or THE, enabling source-wise comparison later in the dashboards.
- 4. Data Integration** — The QS and THE datasets were integrated on common fields (university, country, rank, score) to build a unified working dataset.
- 5. Missing-Value Handling** — Missing values across ranking, score, and related fields were identified and handled during preprocessing to maintain data quality for visualization.
- 6. Data Type Conversion** — Rank and score fields, originally stored in mixed or text formats, were converted into consistent numeric types.
- 7. Rank Processing** — Rank_Numeric and Rank_Normalized fields were created to make ranks comparable and usable in Tableau's rank-based filters and sorts.
- 8. Score Processing** — Overall_Score_Numeric was created from the raw score fields to standardize the overall performance measure used across dashboards.
- 9. Text Standardization** — University and Country text values were standardized (consistent naming/casing) to avoid duplicate groupings caused by inconsistent spelling or formatting.
- 10. Final Cleaned Dataset** — After cleaning, integration, and standardization, the final dataset was exported and connected to Tableau for dashboard development.

4.1 Preprocessing Workflow Diagram



4.2 Outcome of the Cleaning Process

The preprocessing stage converted two independently structured ranking datasets into a single, analysis-ready table. This cleaned dataset formed the direct input for every KPI, chart, and filter used across the five EduVision Analytics dashboards described later in this document.

- Consistent numeric fields (Rank_Numeric, Rank_Normalized, Overall_Score_Numeric) usable directly in Tableau calculations.
- A unified Source field distinguishing QS-origin and THE-origin records for cross-source comparison.
- Standardized University and Country text values, avoiding duplicate groupings in dashboard filters and charts.
- A single cleaned dataset file used as the sole data source for all five Tableau dashboards.

5. Data Analytics & KPI Methodology

After preprocessing, the cleaned dataset was analyzed to build the key performance indicators (KPIs) and comparative visualizations used throughout the five EduVision Analytics dashboards.

5.1 Analysis Areas

Analysis Area	Focus
Ranking Analysis	Uses Rank_Numeric and Rank_Normalized to study university standing and rank-based distribution.
Overall Score Analysis	Uses Overall_Score_Numeric to compare universities on a standardized performance scale.
Research Analysis	Examines research-related indicators such as citations, research productivity and research environment.
Student Analysis	Examines faculty support, teaching strength and student experience indicators.
Country Comparison	Aggregates university-level data by country to compare academic standing, faculty balance and outlook.

5.2 KPI Calculation Concepts

- **Normalization** — Rank and score fields from QS and THE are brought onto comparable scales (Rank_Normalized, Overall_Score_Numeric) so records from both sources can be analyzed together.
- **Aggregation** — KPI cards such as Total Universities, Overall Score, Average Global Ranking and Average Research Score are computed by aggregating the cleaned fields across the dataset.
- **Performance Categorization** — Universities and countries are grouped for comparison (e.g. Top 5 Universities, Top 10 Country Performance, Research Elite) to highlight leading performers within each dashboard.
- **Cross-Source Comparison** — The Source field allows KPIs and charts (e.g. Source Score Spread, Source Insights) to be compared between QS-origin and THE-origin records.

5.3 Summary of Home Dashboard KPIs

KPI	Purpose
Total Universities	Count of unique universities present in the cleaned dataset.
Overall Score	Aggregated overall performance score across all universities.
Average Global Ranking	Average of Rank_Numeric across the dataset.
Average Research Score	Average research-related score across universities.

6. Tableau Dashboard Design & Analytics Approach

The cleaned dataset was connected to Tableau to design five interactive dashboards — Home, University, Research, Student, and Country Comparison. Each dashboard was built around a consistent visual language (KPI cards at the top, filters on the side, and a grid of supporting charts) so that users can move between dashboards without losing context.

6.1 Tableau Visualization Workflow

- Connect the final cleaned dataset (output of the Python preprocessing stage) to Tableau.
- Build calculated fields required for KPI cards and chart measures.
- Design KPI cards for the headline metrics of each dashboard.
- Build supporting charts — bar charts, scatter plots, line/trend charts, tree maps, pie/donut charts, and maps.
- Add interactive filters (e.g. Rank, Tier, Country, Source, University, Learning Index, Geographic View).
- Link navigation buttons across the top of every dashboard (Home, University, Research, Student, Country) for seamless movement between views.
- Test interactivity so that KPI cards and charts update correctly when filters are applied.

6.2 Common Dashboard Structure

Element	Role in the Dashboard
KPI Cards	Display headline numeric indicators at the top of each dashboard for a quick summary.
Filters Panel	Located on the left; lets users narrow the view by rank, tier, country, source or university.
Charts Grid	A grid of bar, scatter, line, tree map and distribution charts supporting deeper analysis.
Maps	Used in the Country Comparison dashboard to show geographic distribution of performance.
Navigation Bar	Top navigation (Home / University / Research / Student / Country) enables movement between all five dashboards.

6.3 Overall Analytics Workflow

QS + THE Datasets	→	Python Cleaning (Jupyter Notebook)	→	Cleaned & Standardized Data	→	Tableau Dashboard Development	→	EduVision Analytics (5 Dashboards)
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The five dashboards were designed to be explored either independently (for a focused view of one dimension of performance) or together (using the shared navigation bar) for a complete analysis of global higher-education performance.

7. Home & University Dashboard

This page presents the two entry-point dashboards of EduVision Analytics — the Home dashboard, which gives an overall summary of global university performance, and the University dashboard, which focuses on detailed, university-level comparison.

7.1 Home Dashboard

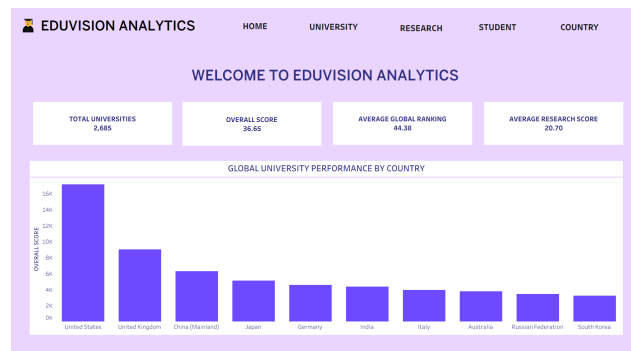


Figure 1: Home Dashboard — EduVision Analytics

- **KPIs:** Total Universities, Overall Score, Average Global Ranking, Average Research Score.
- **Main visualization:** Global University Performance by Country — a bar chart showing performance across the top countries such as the United States, United Kingdom, China (Mainland), Japan and Germany.
- **Purpose:** Gives a single-glance summary of the full dataset before users move into the more detailed dashboards.

7.2 University Dashboard

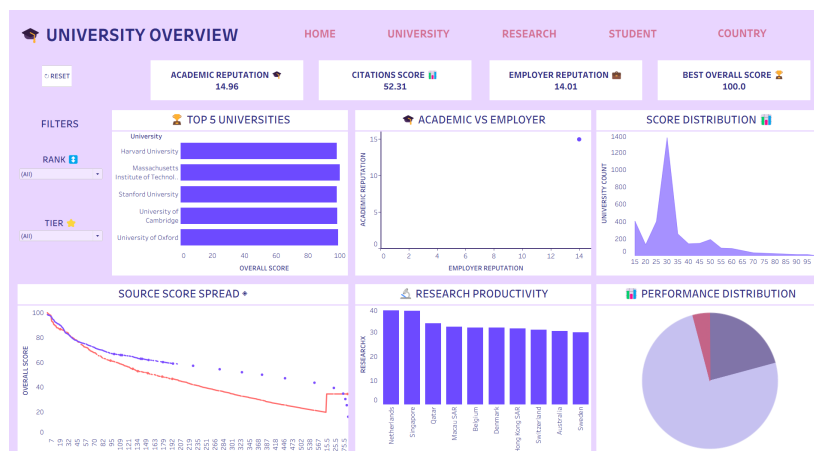


Figure 2: University Dashboard — EduVision Analytics

- **KPIs:** Academic Reputation, Citations Score, Employer Reputation, Best Overall Score.
- **Top 5 Universities:** Ranks the leading universities in the filtered view by overall score.
- **Academic vs Employer:** Scatter analysis comparing academic reputation against employer reputation.
- **Score Distribution:** Area chart showing how universities are distributed across score ranges.
- **Source Score Spread:** Compares score trends between the QS and THE sources.
- **Research Productivity & Performance Distribution:** Bar and pie visualizations summarizing research output and overall score share by country/category.
- **Filters:** Rank and Tier filters let users narrow the analysis to specific rank bands or performance tiers.

8. Research Dashboard

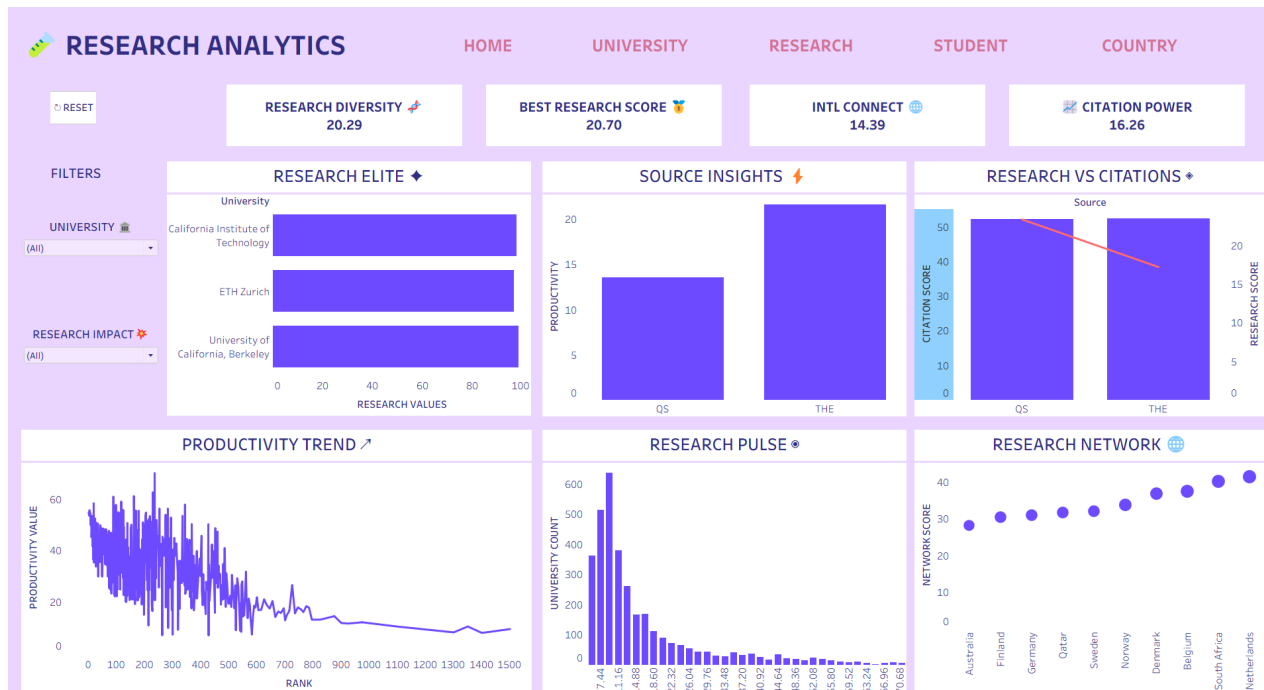


Figure 3: Research Dashboard — EduVision Analytics

The Research dashboard focuses on the research strength of universities, using citation and productivity indicators derived from the cleaned QS and THE data.

- **Research Diversity, Best Research Score, Intl Connect, Citation Power:** KPI cards summarizing the overall research profile of the filtered universities.
- **Research Elite:** Highlights the top research-performing universities such as California Institute of Technology, ETH Zurich and University of California, Berkeley.
- **Source Insights:** Compares research productivity values between the QS and THE sources.
- **Research vs Citations:** Dual-axis bar/line chart comparing citation score against research score by source.
- **Productivity Trend:** Line chart showing how productivity values change across the ranking range.
- **Research Pulse:** Histogram showing how university counts are distributed across productivity bands.
- **Research Network:** Scatter chart plotting network score by country (e.g. Australia, Finland, Germany, Sweden, Netherlands), indicating international research connectivity.
- **Filters:** University and Research Impact filters allow focused analysis on specific institutions or impact levels.

Together, these visualizations help identify which universities and countries lead in research output, citation strength and international research connectivity.

9. Student Dashboard

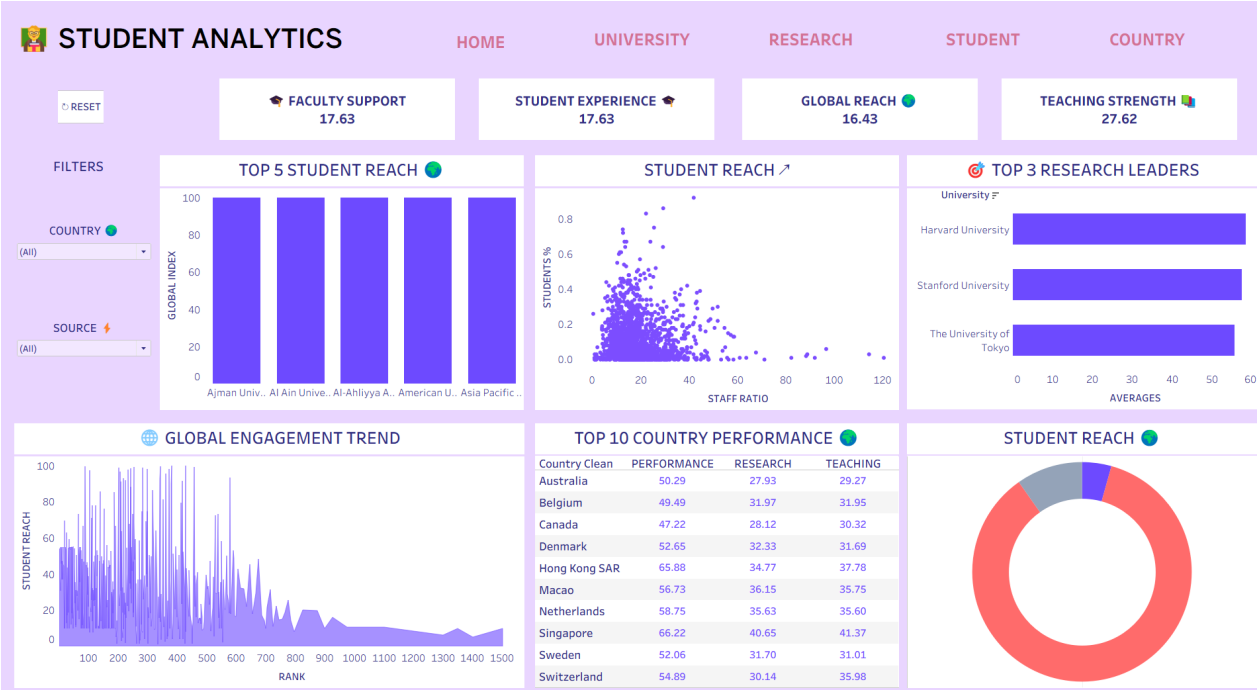


Figure 4: Student Dashboard — EduVision Analytics

The Student dashboard focuses on the aspects of higher-education performance that most directly relate to students — faculty support, teaching strength, and global engagement.

- **Faculty Support, Student Experience, Global Reach, Teaching Strength:** KPI cards summarizing the student-facing performance measures of the filtered universities.
- **Top 5 Student Reach:** Bar chart ranking universities by their global index / student reach.
- **Student Reach (scatter):** Plots students % against staff ratio to show the relationship between faculty availability and student population.
- **Top 3 Research Leaders:** Highlights leading universities (e.g. Harvard University, Stanford University, The University of Tokyo) by average research-related performance.
- **Global Engagement Trend:** Line/area chart showing student reach trend across the ranking range.
- **Top 10 Country Performance:** Table comparing performance, research and teaching scores by country (e.g. Australia, Belgium, Canada, Denmark, Hong Kong SAR, Macao, Netherlands, Singapore, Sweden, Switzerland).
- **Student Reach (donut chart):** Shows the proportional share of student reach across categories.
- **Filters:** Country and Source filters allow the student-related data to be explored by geography or ranking source.

10. Country Comparison Dashboard

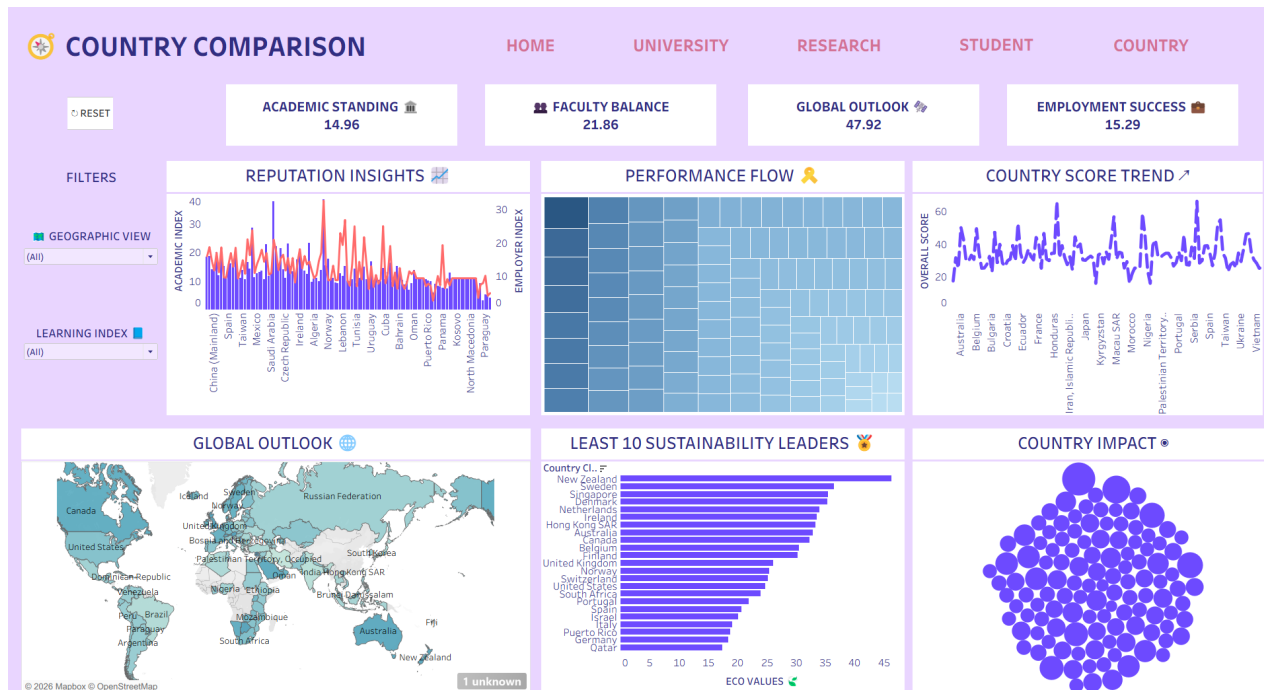


Figure 5: Country Comparison Dashboard — EduVision Analytics

The Country Comparison dashboard aggregates university-level data to the country level, allowing performance to be compared geographically rather than institution by institution.

- **Academic Standing, Faculty Balance, Global Outlook, Employment Success:** KPI cards summarizing country-level academic and employment-related performance.
- **Reputation Insights:** Bar/line combination comparing academic index and employer index across countries.
- **Performance Flow:** Tree-map style visualization showing the relative performance share of different countries.
- **Country Score Trend:** Line chart showing overall score trend across a broad set of countries.
- **Global Outlook Map:** World map showing the geographic distribution of universities and their performance.
- **Least/Top Sustainability Leaders:** Bar chart comparing countries on eco/sustainability-related values, including countries such as New Zealand, Sweden, Singapore, Denmark and the Netherlands.
- **Country Impact:** Bubble visualization representing the relative impact/weight of each country in the dataset.
- **Filters:** Geographic View and Learning Index filters let users focus the comparison on specific regions or learning-related categories.

11. Conclusion, Future Scope & GitHub

11.1 Conclusion

EduVision Analytics successfully transforms QS and Times Higher Education ranking data into an interactive, multi-dashboard analytics solution. The project covers the complete pipeline from data collection through Python-based cleaning and standardization to Tableau dashboard development, resulting in five focused dashboards — Home, University, Research, Student, and Country Comparison — that together allow global higher-education performance to be explored from multiple angles.

11.2 Major Outcomes

- A cleaned and standardized dataset integrating QS and THE ranking data, with consistent rank and score fields.
- Five interactive Tableau dashboards covering overall performance, university-level, research, student, and country-level analysis.
- KPI cards, comparative charts, and a global map that together support both broad and focused exploration of the data.
- A documented, reproducible workflow from raw datasets to final dashboards, suitable for academic submission and future extension.

11.3 Future Scope

- **Predictive Analytics:** Extending the project with forecasting models to project future ranking or score trends.
- **Additional Ranking Datasets:** Incorporating other ranking sources (e.g. ARWU/Shanghai Ranking) for wider comparison.
- **Interactive Improvements:** Adding further drill-down interactivity and cross-dashboard filter actions.
- **Deployment & Year-wise Comparison:** Publishing the dashboards on Tableau Public, and adding historical QS/THE data to analyze how university performance changes over time.

11.4 GitHub Repository

The complete project — including the datasets, Python/Jupyter preprocessing notebook, the cleaned dataset, the Tableau workbook, and dashboard screenshots — is maintained in the following GitHub repository:

github.com/springboardmentorpl/Higher-Education-Performance-Dashboard-Group-1/tree/UnniKrishnan_R

11.5 Key Learning Outcomes

- Practical experience integrating and reconciling two independently structured real-world datasets (QS and THE).
- Hands-on application of Python and Pandas for data cleaning, standardization, and preprocessing in Jupyter Notebook.
- Design of KPI-driven, multi-dashboard analytics using Tableau, including filters, navigation, and comparative visualizations.

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